Orthogonality, Gram–Schmidt, and Orthogonal Matrices

Linear Algebra · Lecture 6

Last time (Lecture 5): block matrices and determinants; $\det(AB) = \det A \det B$, $\det A' = \det A$, and $A \in \mathbb{R}^{n\times n}$ is invertible $\iff$ $\det A \neq 0$. Throughout, $I = I_n$ and e_j is column j of I.

From Lecture 1, Lecture 2, Lecture 3, Lecture 4 and Lecture 5 (restated).

- Lecture 1, Definition 7, Theorems 3, 8: $(A')_{ij} = a_{ji}$; $(AB)C = A(BC)$, $A(B+C) = AB + AC$, $(A+B)C = AC+BC$, $c(AB) = (cA)B = A(cB)$, $I_m A = A = AI_n$ ($A \in \mathbb{R}^{m\times n}$); $(A+B)' = A'+B'$, $(AB)' = B'A'$, $(A')' = A$, $I' = I$; $x'y = y'x = x_1y_1 + \cdots + x_ny_n$ ($x, y \in \mathbb{R}^n$).
- Lecture 1, Definition 1, Theorem 4: for $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, $\operatorname{col}_j(AB) = A\operatorname{col}_j(B)$; $Ax = x_1a_1 + \cdots + x_na_n$, $Ae_j = a_j$; $(AB)_{ij} = \operatorname{row}_i(A)\operatorname{col}_j(B)$, so $(A'G)_{ij} = a_i' \operatorname{col}_j(G)$ ($G \in \mathbb{R}^{m\times p}$); $AB = a_1 \operatorname{row}_1(B) + \cdots + a_n \operatorname{row}_n(B)$. Matrices with the same columns are equal.
- Lecture 1, Definition 14, Example 16: $\mathbf{1} = \mathbf{1}_n = (1, \ldots, 1)' \in \mathbb{R}^n$, $\bar{y} = \frac{1}{n}\mathbf{1}'y$, $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$, $C_n = I_n - \bar{J}_n$; $\sum_i (y_i - \bar{y})^2 = \sum_i y_i^2 - n\bar{y}^2$.
- Lecture 2, Theorems 11(b), 15: for invertible $A, B \in \mathbb{R}^{n\times n}$, AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; for $A, B \in \mathbb{R}^{n\times n}$, $BA = I$ or $AB = I \Rightarrow A$ is invertible and $A^{-1} = B$.
- Lecture 3, Definition 6, Theorem 7, Proposition 12(a),(c): $\operatorname{span}\{v_1, \ldots, v_k\} = \{c_1v_1 + \cdots + c_kv_k : c_i \in \mathbb{R}\}$ is a subspace that contains $v_1, \ldots, v_k$ and lies in every subspace containing them. A list containing 0 is dependent; for $k \geqslant 2$, $v_1, \ldots, v_k$ are dependent $\iff$ some v_j is a combination of the others.
- Lecture 3, Definition 18, Theorems 19(a), 20, 21: for subspaces U, W of $\mathbb{R}^n$, $U + W = \{u + w : u \in U,\ w \in W\}$ is a subspace. If $U \cap W = \{0\}$, the sum is **direct**, written $U \oplus W$; each of its vectors is $u + w$ ($u \in U$, $w \in W$) in exactly one way, and a basis of U together with a basis of W is a basis of $U \oplus W$.
- Lecture 4, Theorem 4: every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements, $\dim W$; $\dim \mathbb{R}^n = n$, $\dim\{0\} = 0$ (empty list), and $\dim W = 0$ only for $W = \{0\}$. If W is spanned by m vectors, then any $k > m$ vectors of W are linearly dependent.
- Lecture 4, Theorem 5(a),(b),(d): let $U \subseteq W$ be subspaces of $\mathbb{R}^n$, $\dim W = d$. Then $\dim U = d \Rightarrow U = W$; an independent list in W extends to a basis of W; d independent vectors of W form a basis of W.
- Lecture 4, Definitions 7, 10, Theorems 11, 12(b): for $A \in \mathbb{R}^{m\times n}$, $\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\}$ (the span of the columns of A) and $\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}$ are subspaces; $\operatorname{rank} A = \dim \mathcal{C}(A)$; $\operatorname{rank} A + \dim \mathcal{N}(A') = m$; $\operatorname{rank} A = n \iff$ the columns of A are independent.
- Lecture 5, Theorems 8(d), 12(b), 13(b): for $A, B \in \mathbb{R}^{n\times n}$, $\det I = 1$, $\det(AB) = \det A \det B$, $\det A' = \det A$.

1. Inner products, lengths, and orthogonal complements

Definition 1. For $x, y \in \mathbb{R}^n$: the **inner product** $x'y = x_1y_1 + \cdots + x_ny_n = y'x$; the **length** $\|x\| = \sqrt{x'x}$; the **distance** $\|x - y\|$; $x \perp y$ (**orthogonal**) if $x'y = 0$. As $x'x$ is a sum of squares, $\|x\| \geqslant 0$, and $\|x\| = 0$ only for $x = 0$: 0 is the only vector orthogonal to itself. $\|cx\| = |c|\,\|x\|$ ($c \in \mathbb{R}$; $(cx)'(cx) = c^2x'x$). A **unit vector** has length 1; for $x \neq 0$, $x/\|x\|$ is one.

Theorem 2. For $x, y \in \mathbb{R}^n$: $\|x+y\|^2 = \|x\|^2 + 2x'y + \|y\|^2$. So $x \perp y \Rightarrow \|x+y\|^2 = \|x\|^2 + \|y\|^2$ (**Pythagoras**).

Proof.

Expand (Lecture 1, Theorems 3, 8):

$$\|x+y\|^2 = (x+y)'(x+y) = x'x + x'y + y'x + y'y = \|x\|^2 + 2x'y + \|y\|^2$$

□

Theorem 3. For $x, y \in \mathbb{R}^n$: $|x'y| \leqslant \|x\|\,\|y\|$ (**Cauchy–Schwarz inequality**), with equality $\iff$ one of x, y is a multiple of the other.

Proof. $x = 0$: both sides are 0, and $x = 0y$. Let $x \neq 0$, $c = x'y/\|x\|^2$ and $w = y - cx$.

Inequality:

$$\begin{aligned}
(cx)'w = c(x'y - c\,x'x) = c(x'y - x'y) = 0 && (x'x = \|x\|^2)\\
\|y\|^2 = \|cx + w\|^2 = c^2\|x\|^2 + \|w\|^2 \geqslant c^2\|x\|^2 = (x'y)^2/\|x\|^2 && \text{(Theorem 2)}\\
(x'y)^2 \leqslant \|x\|^2\|y\|^2 \implies |x'y| \leqslant \|x\|\,\|y\| && \text{(times } \|x\|^2 > 0\text{; square roots)}
\end{aligned}$$

Equality:

$$\begin{aligned}
\text{equality} \iff \|w\|^2 = 0 \iff w = 0 \iff y = cx && \text{(Definition 1)}\\
y = ax \implies c = a\,x'x/\|x\|^2 = a,\ w = 0 &&\\
x = ay,\ x \neq 0 \implies a \neq 0,\ y = a^{-1}x &&
\end{aligned}$$

□

Definition 4. Let W be a subspace of $\mathbb{R}^n$. $y \in \mathbb{R}^n$ is **orthogonal to** W, $y \perp W$, if $y'w = 0$ for every $w \in W$. $W^\perp = \{y \in \mathbb{R}^n : y \perp W\}$ is the **orthogonal complement** of W.

Theorem 5. Let W be a subspace of $\mathbb{R}^n$.

(a) If $W = \operatorname{span}\{w_1, \ldots, w_k\}$ and $A = [w_1 \ \cdots \ w_k]$: $y \perp W \iff y'w_1 = \cdots = y'w_k = 0 \iff A'y = 0$. So $W^\perp = \mathcal{N}(A')$, a subspace of $\mathbb{R}^n$.

(b) $W \cap W^\perp = \{0\}$.

(c) $\dim W + \dim W^\perp = n$, and $\mathbb{R}^n = W \oplus W^\perp$: every $y \in \mathbb{R}^n$ is $y = w + v$ with $w \in W$, $v \in W^\perp$, in exactly one way.

Proof.

(a) The entries of $A'y$ are $w_j'y$; $w \in W$ is $c_1w_1 + \cdots + c_kw_k = Ac$ (Lecture 1, Theorems 4, 8; Lecture 3, Definition 6):

$$\begin{aligned}
y \perp W \implies y'w_j = w_j'y = 0\ (j = 1, \ldots, k) \iff A'y = 0 && (w_j \in W)\\
A'y = 0,\ w = Ac \in W \implies y'w = y'Ac = (A'y)'c = 0 &&
\end{aligned}$$

So $W^\perp = \mathcal{N}(A')$, a subspace (Lecture 4, Definition 7).

(b) Let $y \in W \cap W^\perp$:

$$y \in W,\ y \perp W \implies y'y = 0 \implies y = 0 \qquad \text{(Definition 1)}$$

(c) Let $d = \dim W$. If $d = 0$: $W = \{0\}$, $W^\perp = \mathbb{R}^n$. Let $d \geqslant 1$, and let $A \in \mathbb{R}^{n \times d}$ have a basis of W as its columns (Lecture 4, Theorem 4).

(c) $\dim W^\perp$ (Lecture 4, Definitions 7, 10, Theorem 11, with $m = n$):

$$\mathcal{C}(A) = W, \quad \operatorname{rank} A = \dim W = d$$
$$\dim W^\perp = \dim \mathcal{N}(A') = n - \operatorname{rank} A = n - d \qquad ((a))$$

(c) $\mathbb{R}^n = W \oplus W^\perp$ (Lecture 3, Definition 18, Theorems 19(a), 20, 21; Lecture 4, Theorems 4, 5(a)):

$$W \cap W^\perp = \{0\} \implies W + W^\perp = W \oplus W^\perp, \text{ a subspace of } \mathbb{R}^n \qquad ((b))$$
bases of W, $W^\perp$ together: $d + (n - d) = n$ vectors, a basis of $W \oplus W^\perp$
$$\dim(W \oplus W^\perp) = n = \dim \mathbb{R}^n \implies W \oplus W^\perp = \mathbb{R}^n$$

Each $y \in \mathbb{R}^n$ is $w + v$ ($w \in W$, $v \in W^\perp$) in exactly one way.

□

Corollary 6. For $A \in \mathbb{R}^{m\times n}$: $\mathcal{N}(A) = \mathcal{C}(A')^\perp$, $\mathbb{R}^n = \mathcal{C}(A') \oplus \mathcal{N}(A)$; $\mathcal{N}(A') = \mathcal{C}(A)^\perp$, $\mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A')$.

Proof.

$\mathcal{C}(A')$, $\mathcal{C}(A)$ are spanned by the columns of A', A (Lecture 4, Definition 7):

$$\mathcal{C}(A')^\perp = \mathcal{N}\big((A')'\big) = \mathcal{N}(A), \qquad \mathcal{C}(A)^\perp = \mathcal{N}(A') \qquad \text{(Theorem 5(a); Lecture 1, Theorem 8)}$$
$$\mathbb{R}^n = \mathcal{C}(A') \oplus \mathcal{C}(A')^\perp, \qquad \mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{C}(A)^\perp \qquad \text{(Theorem 5(c))}$$

□

2. Orthonormal bases, Gram–Schmidt, and QR

Theorem 7. Let $u_1, \ldots, u_k \in \mathbb{R}^n$ be nonzero, with $u_i' u_j = 0$ for $i \neq j$.

(a) If $y = c_1 u_1 + \cdots + c_k u_k$, then $c_j = u_j' y / \|u_j\|^2$ ($j = 1, \ldots, k$).

(b) $u_1, \ldots, u_k$ are linearly independent.

Proof.

(a)

$$u_j' y = c_1 u_j' u_1 + \cdots + c_k u_j' u_k = c_j \|u_j\|^2, \quad \|u_j\|^2 > 0 \qquad \text{(Lecture 1, Theorem 3; Definition 1)}$$

(b)

$$c_1 u_1 + \cdots + c_k u_k = 0 \implies c_j = u_j' 0 / \|u_j\|^2 = 0 \ (j = 1, \ldots, k) \qquad ((a),\ y = 0)$$

□

Definition 8. $q_1, \ldots, q_p \in \mathbb{R}^n$ are **orthonormal** if $q_i' q_j = 0$ ($i \neq j$) and $\|q_j\| = 1$ (all j). $Q = [q_1 \ \cdots \ q_p] \in \mathbb{R}^{n\times p}$ has **orthonormal columns** if $Q'Q = I_p$, that is, $q_1, \ldots, q_p$ are orthonormal, as $(Q'Q)_{ij} = q_i' q_j$ (Lecture 1, Theorem 4). $Q \in \mathbb{R}^{n\times n}$ with $Q'Q = I_n$ is an **orthogonal matrix**. An **orthonormal basis** of a subspace is a basis of orthonormal vectors.

Theorem 9. Let $Q \in \mathbb{R}^{n\times p}$, $Q'Q = I_p$.

(a) $(Qx)'(Qy) = x'y$ and $\|Qx\| = \|x\|$ $(x, y \in \mathbb{R}^p)$.

(b) If $p = n$: Q is invertible, $Q^{-1} = Q'$ and $QQ' = I_n$.

Proof.

(a)

$$(Qx)'(Qy) = x'Q'Qy = x'I_py = x'y; \qquad y = x\colon\ \|Qx\|^2 = \|x\|^2 \qquad \text{(Lecture 1, Theorems 3, 8)}$$

(b) Q is square:

$$Q'Q = I_n \implies Q \text{ invertible, } Q^{-1} = Q', \quad QQ' = QQ^{-1} = I_n \qquad \text{(Lecture 2, Theorem 15)}$$

□

Theorem 10 (Gram–Schmidt)**.** Let $a_1, \ldots, a_p \in \mathbb{R}^n$ be linearly independent. Let $b_1 = a_1$ and, for $j = 2, \ldots, p$,

$$b_j = a_j - t_{1j}b_1 - t_{2j}b_2 - \cdots - t_{j-1,j}b_{j-1}, \qquad t_{ij} = \frac{a_j'b_i}{\|b_i\|^2} \quad (i < j)$$

Then $b_1, \ldots, b_p$ are nonzero and pairwise orthogonal, and $\operatorname{span}\{b_1, \ldots, b_j\} = \operatorname{span}\{a_1, \ldots, a_j\}$ $(j = 1, \ldots, p)$. So $q_j = b_j/\|b_j\|$ $(j = 1, \ldots, p)$ form an orthonormal basis of $\operatorname{span}\{a_1, \ldots, a_p\}$.

Proof. Induction on j. $b_1 = a_1 \neq 0$ (Lecture 3, Proposition 12(a)). Let $j \geqslant 2$, and let $b_1, \ldots, b_{j-1}$ be nonzero, pairwise orthogonal, with span $W_{j-1} = \operatorname{span}\{a_1, \ldots, a_{j-1}\}$; so b_j is defined.

Orthogonal: for $l < j$, with $b_l'b_i = 0$ $(i \neq l)$:

$$b_l'b_j = b_l'a_j - \textstyle\sum_{i<j} t_{ij}b_l'b_i = b_l'a_j - t_{lj}\|b_l\|^2 = b_l'a_j - a_j'b_l = 0 \qquad \text{(Lecture 1, Theorems 3, 8)}$$

Nonzero:

$$\begin{aligned} b_j = 0 &\implies a_j = \textstyle\sum_{i<j} t_{ij}b_i \in W_{j-1} = \operatorname{span}\{a_1, \ldots, a_{j-1}\} \\ &\implies a_1, \ldots, a_p \text{ dependent } \text{✘} \qquad \text{(Lecture 3, Proposition 12(c))} \end{aligned}$$

Span (Lecture 3, Theorem 7: a span lies in every subspace containing the list):

$$\begin{aligned} b_1, \ldots, b_{j-1} \in W_{j-1} \subseteq \operatorname{span}\{a_1, \ldots, a_j\}, &\quad b_j = a_j - \textstyle\sum_{i<j} t_{ij}b_i \in \operatorname{span}\{a_1, \ldots, a_j\} \\ a_1, \ldots, a_{j-1} \in W_{j-1} \subseteq \operatorname{span}\{b_1, \ldots, b_j\}, &\quad a_j = b_j + \textstyle\sum_{i<j} t_{ij}b_i \in \operatorname{span}\{b_1, \ldots, b_j\} \end{aligned}$$

Each list lies in the span of the other, so the two spans are equal.

Last sentence: the q_j are pairwise orthogonal unit vectors (Definition 1), so independent (Theorem 7(b)), and $\operatorname{span}\{q_1, \ldots, q_p\} = \operatorname{span}\{b_1, \ldots, b_p\} = \operatorname{span}\{a_1, \ldots, a_p\}$ (Lecture 3, Theorem 7). □

Corollary 11. Let W be a subspace of $\mathbb{R}^n$, $\dim W = d \geqslant 1$.

(a) W has an orthonormal basis.

(b) If $q_1, \dots, q_p \in W$ are orthonormal and $p < d$, there are $q_{p+1}, \dots, q_d \in W$ such that $q_1, \dots, q_d$ form an orthonormal basis of W.

(c) If $Q \in \mathbb{R}^{n\times p}$, $Q'Q = I_p$ and $p < n$, then $[Q\ T]$ is orthogonal for some $T \in \mathbb{R}^{n\times(n-p)}$. For every such T, the columns of T form an orthonormal basis of $\mathcal{C}(Q)^\perp$, and $QQ' + TT' = I_n$.

Proof. (a) W has a basis $a_1, \dots, a_d$ (Lecture 4, Theorem 4); Theorem 10 turns it into an orthonormal basis of $\operatorname{span}\{a_1, \dots, a_d\} = W$.

(b) $q_1, \dots, q_p$ are independent (Theorem 7(b)); extend (Lecture 4, Theorem 5(b)), then Gram–Schmidt (Theorem 10):

$$\begin{aligned}
&q_1, \dots, q_p, a_{p+1}, \dots, a_d\text{: a basis of } W\\
&b_1 = q_1;\quad b_i = q_i\ (i < j \leqslant p) \implies t_{ij} = q_j'q_i/\|q_i\|^2 = 0 \implies b_j = q_j\\
&q_1, \dots, q_p,\ b_{p+1}/\|b_{p+1}\|, \dots, b_d/\|b_d\|\text{: an orthonormal basis of } W
\end{aligned}$$

(c) T exists: the columns $q_1, \dots, q_p$ of Q are orthonormal (Definition 8); $\dim \mathbb{R}^n = n$ (Lecture 4, Theorem 4):

$$\begin{aligned}
&p < n \implies q_{p+1}, \dots, q_n \text{ with } q_1, \dots, q_n \text{ an orthonormal basis of } \mathbb{R}^n && ((b),\ W = \mathbb{R}^n)\\
&T = [q_{p+1}\ \cdots\ q_n] \implies [Q\ T] \text{ has orthonormal columns: orthogonal} && (\text{Definition 8})
\end{aligned}$$

(c) Let $T = [v_1\ \cdots\ v_{n-p}]$ be any such matrix: $[Q\ T]'[Q\ T] = I_n$, so $q_1, \dots, q_p, v_1, \dots, v_{n-p}$ are orthonormal (Definition 8), hence independent (Theorem 7(b)); $\mathcal{C}(Q) = \operatorname{span}\{q_1, \dots, q_p\}$ (Lecture 4, Definition 7):

$$\begin{aligned}
v_k'q_i = 0\ (\text{all } k,\ i) &\implies v_k \in \mathcal{C}(Q)^\perp && (\text{Theorem 5(a)})\\
\operatorname{rank} Q = p &\implies \dim \mathcal{C}(Q)^\perp = n - p && (\text{Lecture 4, Theorem 12(b); Theorem 5(c)})\\
&\implies v_1, \dots, v_{n-p} \text{ form a basis of } \mathcal{C}(Q)^\perp && (\text{Lecture 4, Theorem 5(d)})\\
I_n &= [Q\ T][Q\ T]' && (\text{Theorem 9(b)})\\
&= \textstyle\sum_i q_iq_i' + \sum_k v_kv_k' = QQ' + TT' && (\text{Lecture 1, Theorem 4})
\end{aligned}$$

□

Theorem 12 (QR decomposition)**.** Let $A = [a_1\ \cdots\ a_p] \in \mathbb{R}^{n\times p}$, $\operatorname{rank} A = p$. Then $A = QR$ with $Q \in \mathbb{R}^{n\times p}$, $Q'Q = I_p$, and $R = (r_{ij}) \in \mathbb{R}^{p\times p}$ upper triangular ($r_{ij} = 0$ for $i > j$) with $r_{jj} > 0$; moreover $R = Q'A$.

Proof.

$a_1, \dots, a_p$ are independent (Lecture 4, Theorem 12(b)); Theorem 10 gives b_j, t_{ij}, q_j; $Q = [q_1\ \cdots\ q_p]$, $Q'Q = I_p$ (Definition 8):

$$\begin{aligned}
r_{jj} = \|b_j\| > 0,\quad r_{ij} = t_{ij}\|b_i\|\ (i < j),\quad r_{ij} = 0\ (i > j) &\quad (b_j \neq 0,\ \text{Theorem 10; Definition 1})\\
a_j = b_j + \textstyle\sum_{i<j} t_{ij}b_i = \sum_{i\leqslant j} r_{ij}q_i &\quad (b_i = \|b_i\|q_i)\\
\operatorname{col}_j(QR) = Q\operatorname{col}_j(R) = \textstyle\sum_{i\leqslant j} r_{ij}q_i = a_j\ (\text{all } j) \implies QR = A &\quad (\text{Lecture 1, Theorem 4})\\
Q'A = Q'QR = I_pR = R &\quad (\text{Lecture 1, Theorem 3})
\end{aligned}$$

□

Example 13. Find the QR decomposition of

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 0 & 2 & 4 \\ 2 & 2 & 5 \end{pmatrix}$$

Solution.

Gram–Schmidt (Theorem 10) on the columns a_1, a_2, a_3:

$$\begin{aligned} b_1 &= a_1 = (1,2,0,2)', \quad \|b_1\|^2 = 1+4+0+4 = 9 \\ a_2'b_1 &= 3+2+0+4 = 9, \quad t_{12} = 1 \\ b_2 &= a_2 - b_1 = (3,1,2,2)' - (1,2,0,2)' = (2,-1,2,0)', \quad \|b_2\|^2 = 4+1+4+0 = 9 \\ a_3'b_1 &= 2+6+0+10 = 18, \quad a_3'b_2 = 4-3+8+0 = 9, \quad t_{13} = 2, \quad t_{23} = 1 \\ b_3 &= a_3 - 2b_1 - b_2 = (2,3,4,5)' - (2,4,0,4)' - (2,-1,2,0)' = (-2,0,2,1)' \\ \|b_3\|^2 &= 4+0+4+1 = 9 \end{aligned}$$

Check: $b_1'b_2 = 2-2+0+0 = 0$, $b_1'b_3 = -2+0+0+2 = 0$, $b_2'b_3 = -4+0+4+0 = 0$.

Q and R (Theorem 12); all three lengths are 3:

$$q_j = \tfrac{1}{3}b_j, \quad r_{jj} = 3, \quad r_{12} = 3t_{12} = 3, \quad r_{13} = 3t_{13} = 6, \quad r_{23} = 3t_{23} = 3$$

$$Q = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 2 & 2 \\ 2 & 0 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 3 & 3 & 6 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}$$

Check $R = Q'A$: for instance $r_{23} = q_2'a_3 = \frac{1}{3}(4-3+8+0) = 3$.

Example 14. Extend the orthonormal columns q_1, q_2, q_3 of Q in Example 13 to an orthonormal basis of $\mathbb{R}^4$.

Solution. We need a unit vector $q_4 \perp q_1, q_2, q_3$ (Definition 8): $q_4 = v/\|v\|$ with $v \neq 0$, $b_1'v = b_2'v = b_3'v = 0$, that is, $v \in \mathcal{N}([b_1\ b_2\ b_3]')$ (Theorem 5(a)).

Solve by substitution:

$v_1 + 2v_2 + 2v_4 = 0, \qquad 2v_1 - v_2 + 2v_3 = 0, \qquad -2v_1 + 2v_3 + v_4 = 0$

third: $v_4 = 2v_1 - 2v_3$; second: $v_2 = 2v_1 + 2v_3$

first: $v_1 + (4v_1 + 4v_3) + (4v_1 - 4v_3) = 9v_1 = 0 \implies v_1 = 0,\ v_2 = 2v_3,\ v_4 = -2v_3$

$v_3 = 1$: $v = (0,2,1,-2)', \quad v'v = 0+4+1+4 = 9, \quad q_4 = \frac{1}{3}v, \qquad [Q\ q_4] = \frac{1}{3}\begin{pmatrix} 1 & 2 & -2 & 0 \\ 2 & -1 & 0 & 2 \\ 0 & 2 & 2 & 1 \\ 2 & 0 & 1 & -2 \end{pmatrix}$

$[Q\ q_4]$ is orthogonal (its columns are orthonormal, Definition 8). By Corollary 11(c), q_4 alone is an orthonormal basis of $\mathcal{C}(Q)^\perp = \mathcal{C}(A)^\perp$ ($\mathcal{C}(Q) = \mathcal{C}(A)$ by Theorem 10), and $QQ' + q_4q_4' = I_4$.

3. Orthogonal matrices

Theorem 15. (a) For $Q \in \mathbb{R}^{n\times n}$ the following are equivalent: (i) $Q'Q = I_n$ (Q is orthogonal); (ii) $QQ' = I_n$; (iii) Q is invertible and $Q^{-1} = Q'$; (iv) the columns of Q are orthonormal; (v) the transposed rows $\mathrm{row}_1(Q)', \dots, \mathrm{row}_n(Q)'$ are orthonormal.

(b) If $P, Q \in \mathbb{R}^{n\times n}$ are orthogonal, so are PQ, Q' and Q^{-1}.

(c) If Q is orthogonal, then $\det Q = 1$ or $\det Q = -1$.

Proof.

(a) $q_1, \dots, q_n$ are the columns of Q; $\mathrm{col}_j(Q') = \mathrm{row}_j(Q)'$ (Lecture 1, Definition 7):

(i) $\Rightarrow$ (ii), (iii)	(Theorem 9(b))
(ii) $\Rightarrow$ (iii): $QQ' = I_n \implies Q$ invertible, $Q^{-1} = Q'$	(Lecture 2, Theorem 15)
(iii) $\Rightarrow$ (i): $Q'Q = Q^{-1}Q = I_n$	
(i) $\Leftrightarrow$ (iv): $(Q'Q)_{ij} = q_i'q_j$	(Lecture 1, Theorem 4; Definition 8)
(ii) $\Leftrightarrow$ (v): $(QQ')_{ij} = \mathrm{row}_i(Q)\,\mathrm{row}_j(Q)'$	(Lecture 1, Theorem 4; Definition 8)

(b) $P^{-1} = P'$, $Q^{-1} = Q'$ ((a)(iii)); Lecture 2, Theorem 11(b), and Lecture 1, Theorem 8:

PQ invertible, $(PQ)^{-1} = Q^{-1}P^{-1} = Q'P' = (PQ)' \implies PQ$ orthogonal	((a)(iii))
$(Q')'Q' = QQ' = I_n \implies Q'$ orthogonal; $\quad Q^{-1} = Q'$	((a)(ii))

(c) Lecture 5, Theorems 8(d), 12(b), 13(b):

$$1 = \det I_n = \det(Q'Q) = \det Q' \det Q = (\det Q)^2 \implies \det Q = \pm 1$$

□

So, for orthogonal Q: $\|Qx - Qy\| = \|Q(x-y)\| = \|x-y\|$ (Theorem 9(a)); Q keeps inner products, lengths and distances.

Example 16. Let $n \geqslant 2$. For $k = 2, \dots, n$ let $u_k = (1, \dots, 1, -(k-1), 0, \dots, 0)' \in \mathbb{R}^n$: $k-1$ entries 1, then $-(k-1)$ in position k, then zeros. The **Helmert matrix** $K_n \in \mathbb{R}^{n\times n}$ has the rows

$$p_1' = \frac{1}{\sqrt{n}}\,\mathbf{1}_n', \quad p_k' = \frac{1}{\sqrt{k(k-1)}}\,u_k' \ (k = 2, \dots, n); \quad K_4 = \begin{pmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} & 0 \\ 1/\sqrt{12} & 1/\sqrt{12} & 1/\sqrt{12} & -3/\sqrt{12} \end{pmatrix}$$

(a) Show that K_n is orthogonal.

(b) For $y \in \mathbb{R}^n$, $p_1'y = \sqrt{n}\,\bar{y}$, and for $k \geqslant 2$, $p_k'y = \big(y_1 + \cdots + y_{k-1} - (k-1)y_k\big)/\sqrt{k(k-1)}$ is a **contrast** (its coefficients add up to 0). Show that $\sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{k=2}^n (p_k'y)^2$.

(c) Check (b) for $n = 4$, $y = (5, 3, 1, 7)'$.

Solution.

(a) By Theorem 15(a), (v)$\Rightarrow$(i), show that $p_1, \dots, p_n$ are orthonormal:

$$\mathbf{1}'u_k = (k-1) - (k-1) = 0 \ (k \geqslant 2) \implies p_1'p_k = 0$$
$$2 \leqslant j < k\text{: } u_j = 0 \text{ outside positions } 1, \dots, j;\ u_k \text{ has 1 there } (j \leqslant k-1)$$
$$\implies u_j'u_k = \mathbf{1}'u_j = 0 \implies p_j'p_k = 0$$
$$\mathbf{1}'\mathbf{1} = n, \quad u_k'u_k = (k-1)\cdot 1 + (k-1)^2 = k(k-1) \implies \|p_1\| = \|p_k\| = 1$$

(b) The entries of $K_n y$ are $p_k' y$ (Lecture 1, Theorem 4); (a) and Theorem 9(a):

$$\begin{aligned} p_1' y &= \tfrac{1}{\sqrt{n}} \mathbf{1}' y = \tfrac{1}{\sqrt{n}}\, n\bar{y} = \sqrt{n}\,\bar{y} \\ \textstyle\sum_i y_i^2 &= \|y\|^2 = \|K_n y\|^2 = n\bar{y}^2 + \textstyle\sum_{k=2}^n (p_k' y)^2 \\ \textstyle\sum_i (y_i - \bar{y})^2 &= \textstyle\sum_i y_i^2 - n\bar{y}^2 = \textstyle\sum_{k=2}^n (p_k' y)^2 \qquad \text{(Lecture 1, Example 16)} \end{aligned}$$

In matrix form: $K_n' K_n = I_n$ reads $p_1 p_1' + \cdots + p_n p_n' = I_n$ (Lecture 1, Theorem 4), and $p_1 p_1' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$; so $p_2 p_2' + \cdots + p_n p_n' = I_n - \bar{J}_n = C_n$.

(c) $n = 4$, $y = (5, 3, 1, 7)'$:

$$\bar{y} = 4, \quad p_1' y = 16/2 = 8 = \sqrt{4} \cdot 4$$

$$p_2' y = \frac{5-3}{\sqrt{2}} = \sqrt{2}, \quad p_3' y = \frac{5+3-2}{\sqrt{6}} = \sqrt{6}, \quad p_4' y = \frac{5+3+1-21}{\sqrt{12}} = -\sqrt{12}$$

$$2 + 6 + 12 = 20 = 1 + 1 + 9 + 9 = \textstyle\sum_i (y_i - 4)^2, \qquad \|y\|^2 = 25 + 9 + 1 + 49 = 84 = 64 + 20$$

Practice. The Exercises and Problems of the textbook, Chapter 6, on inner products and orthogonal complements, Gram–Schmidt and the QR decomposition, and orthogonal matrices.